AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Design basis: examples/sic-buck-400v-200v-5kw/design/design_decisions.json (schema 3, 2026-10-10) — single source of truth; electrical model automation/kicad_tools/sic_buck_control_model.py; generator automation/kicad_tools/generate_sic_schematics.py.
400 V (360-440 V) to 200 V / 25 A / 5 kW / 50 kHz synchronous buck; Wolfspeed C4MS025120K; TMS320F28335 control board on a stacked SELV mezzanine.
Net classes: CTRL_SELV and Default 0.2 mm (35 um copper), CTRL_PWR for the supply rails (1.0 mm tracks, added assumption A-G2); the shared HV patterns are guards and match no net of this SELV board.
Board rules (.kicad_dru): the baseline class-pair rules are carried as guards; the control board must never carry HV copper and its copper never lies above HV copper of the power board (air clearance >= 11 mm, standards review item).
Polarity adaptations A-C6 (UVLO flag inverted to UVLO_N) and A-C7 (raw PG_BIAS_x_N inverted before the AND combiner) use one SN74LV04A: the power board keeps the raw PG polarity on mezzanine pins 30/32/33.
Evidence: native ERC manifests under verification/erc/<run>/run.json; review PDF under review/; this drawing has no CAD evidence beyond those runs. AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED.

12 V entry, TVS and TPS26400 eFuse

FLT_12V_ND
I12V_MOND
V12V_MOND

FLT_12V_N
I12V_MON
V12V_MON

文件: 01_aux_12v.kicad_sch

5 V buck, dual LDO, supervisor and reset

XRS_N

XRS_N

文件: 02_supplies.kicad_sch

TMS320F28335 — all 176 pins

FAULT_N
FLT_12V_N
OTP_N
OVP_N
UVLO_N

FAULT_N
FLT_12V_N
OTP_N
OVP_N
UVLO_N

ADCREFM
ADCREFP
ADCRESEXT
ADC_I12V
ADC_ISENSE
ADC_ISENSE_REF
ADC_TEMP_CTRL
ADC_TEMP_HS
ADC_V12V
ADC_VIN
ADC_VOUT
ARM
BIAS_EN
BOOT84
BOOT85
BOOT86
BOOT87
CLK30
DSP_TRIP_N
EMU0
EMU1
EPWM1A
EPWM1B
ESTOP_N
FAN_TACH
FLT_H_N
FLT_L_N
LED_FAULT
LED_RUN
OC_N
PG_BIAS_H_N
PG_BIAS_L_N
PG_BIAS_S_N
RDY_H
RDY_L
RELAY_ON
RESET_N
SPARE_D1
SPARE_D2
TCK
TDI
TDO
TMS
TRST
UART_RX
UART_TX
XRS_N

ADCREFM
ADCREFP
ADCRESEXT
ADC_I12V
ADC_ISENSE
ADC_ISENSE_REF
ADC_TEMP_CTRL
ADC_TEMP_HS
ADC_V12V
ADC_VIN
ADC_VOUT
ARM
BIAS_EN
BOOT84
BOOT85
BOOT86
BOOT87
CLK30
DSP_TRIP_N
EMU0
EMU1
EPWM1A
EPWM1B
ESTOP_N
FAN_TACH
FLT_H_N
FLT_L_N
LED_FAULT
LED_RUN
OC_N
PG_BIAS_H_N
PG_BIAS_L_N
PG_BIAS_S_N
RDY_H
RDY_L
RELAY_ON
RESET_N
SPARE_D1
SPARE_D2
TCK
TDI
TDO
TMS
TRST
UART_RX
UART_TX
XRS_N

文件: 03_dsp.kicad_sch

DSP per-pin bypass and ADC reference

ADCREFM
ADCREFP

ADCREFM
ADCREFP

ADCRESEXT

ADCRESEXT

文件: 04_dsp_decoupling.kicad_sch

Oscillator, boot straps, JTAG, UART

TDO

TDO

BOOT84
BOOT85
BOOT86
BOOT87
CLK30
EMU0
EMU1
TCK
TDI
TMS
TRST
UART_RX
UART_TX

BOOT84
BOOT85
BOOT86
BOOT87
CLK30
EMU0
EMU1
TCK
TDI
TMS
TRST
UART_RX
UART_TX

文件: 05_clock_debug.kicad_sch

PWM gating, fault latch, combiners, comparators

FLT_12V_N
XRS_N

FLT_12V_N
XRS_N

ARM
DSP_TRIP_N
EPWM1A
EPWM1B
ESTOP_N
FAULT_LATCH_N
FAULT_N
FLT_H_N
FLT_L_N
GATE_EN
OC_N
OTP_N
OTP_REF
OVP_N
PG_BIAS_H_N
PG_BIAS_L_N
PG_BIAS_S_N
PWM_H
PWM_L
RDY_H
RDY_L
TEMP_HS_NTC
UVLO_N
VIN_SENSE
VOUT_SENSE

ARM
DSP_TRIP_N
EPWM1A
EPWM1B
ESTOP_N
FAULT_LATCH_N
FAULT_N
FLT_H_N
FLT_L_N
GATE_EN
OC_N
OTP_N
OTP_REF
OVP_N
PG_BIAS_H_N
PG_BIAS_L_N
PG_BIAS_S_N
PWM_H
PWM_L
RDY_H
RDY_L
TEMP_HS_NTC
UVLO_N
VIN_SENSE
VOUT_SENSE

文件: 06_pwm_fault_logic.kicad_sch

ADC conditioning, NTC readout, monitors

I12V_MON

I12V_MON

ADC_I12V
ADC_ISENSE
ADC_ISENSE_REF
ADC_TEMP_CTRL
ADC_TEMP_HS
ADC_V12V
ADC_VIN
ADC_VOUT
ISENSE
ISENSE_REF
OTP_REF
TEMP_HS_NTC
V12V_MON
VIN_SENSE
VOUT_SENSE

ADC_I12V
ADC_ISENSE
ADC_ISENSE_REF
ADC_TEMP_CTRL
ADC_TEMP_HS
ADC_V12V
ADC_VIN
ADC_VOUT
ISENSE
ISENSE_REF
OTP_REF
TEMP_HS_NTC
V12V_MON
VIN_SENSE
VOUT_SENSE

文件: 07_adc_frontend.kicad_sch

Mezzanine header, connectors, LEDs, test points, mounting

FAULT_LATCH_N
FAULT_N
GATE_EN
XRS_N

FAULT_LATCH_N
FAULT_N
GATE_EN
XRS_N

ARM
BIAS_EN
FAN_TACH
FLT_H_N
FLT_L_N
ISENSE
ISENSE_REF
LED_FAULT
LED_RUN
OC_N
PG_BIAS_H_N
PG_BIAS_L_N
PG_BIAS_S_N
PWM_H
PWM_L
RDY_H
RDY_L
RELAY_ON
RESET_N
SPARE_D1
SPARE_D2
VIN_SENSE
VOUT_SENSE

ARM
BIAS_EN
FAN_TACH
FLT_H_N
FLT_L_N
ISENSE
ISENSE_REF
LED_FAULT
LED_RUN
OC_N
PG_BIAS_H_N
PG_BIAS_L_N
PG_BIAS_S_N
PWM_H
PWM_L
RDY_H
RDY_L
RELAY_ON
RESET_N
SPARE_D1
SPARE_D2
VIN_SENSE
VOUT_SENSE

文件: 08_interconnect.kicad_sch

AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /
File: AIPE_SIC_Buck_Control.kicad_sch

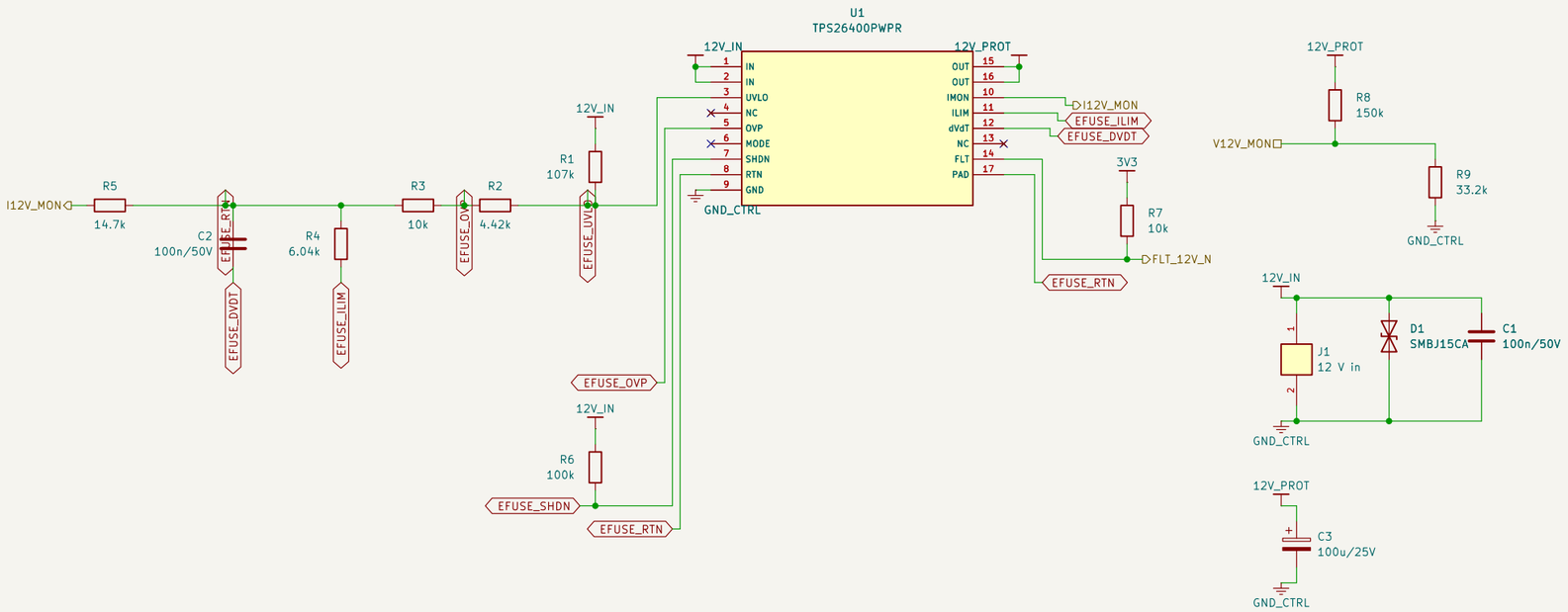
Title: SiC buck 400V / 200V / 5kW — control board

Size: A2 | Date: | **Rev: V2.2**
KiCad E.D.A. 10.0.7 | Id: 1/9

12 V entry, TVS and TPS26400 eFuse

TPS26400 eFuse: UVLO 10.0 / 9.3 V and OVP 14.45 / 13.4 V from 107 k / 4.42 k / 10.0 k, R_ILIM 6.04 k (I_CB about 1.96 A), C_dVdT 100 nF, R_IMON 14.7 k, MODE open = circuit breaker (CR-18). RTN (EFUSE_RTN) is never tied to GND_CTRL (A-C3). PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.

12 V entry, TVS and TPS26400 eFuse



Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /12 V entry, TVS and TPS26400 eFuse/
File: 01_aux_12v.kicad_sch

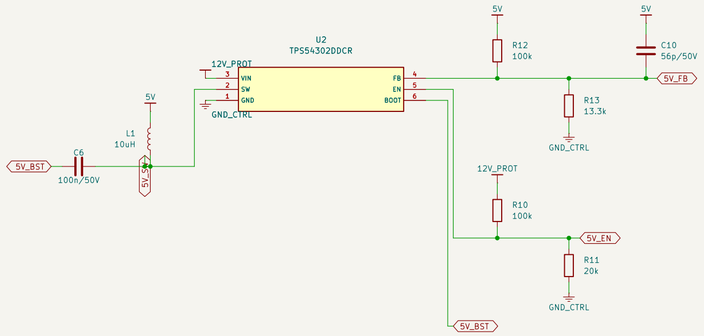
Title: 12 V entry, TVS and TPS26400 eFuse

Size: A3 | Date: | **Rev: V2.2**
KiCad E.D.A. 10.0.7 | Id: 2/9

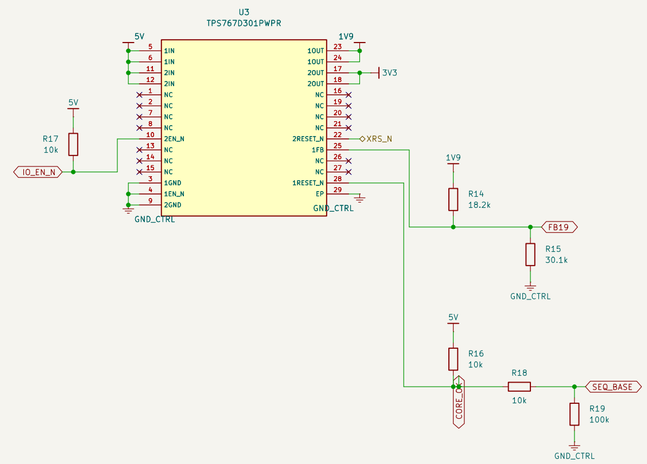
5 V buck, dual LDO, supervisor and reset

TPS54302 5 V buck (FB 100 k / 13.3 k), TPS767D301 dual LDO with core-first sequencing (1.9 V from 18.2 k / 30.1 k, 3.3 V fixed), TPS3808G01 supervisor 101 k / 10 k (4.50 V threshold, CT 10 nF). The -EP LDO variant is NRND; TLV752 is the named alternative.

TPS54302 5 V buck



TPS767D301 dual LDO, core-first sequencing

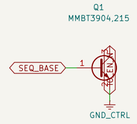


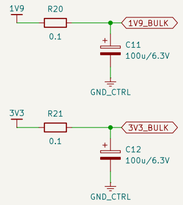
U4 dedicated local bypass (V2.2)

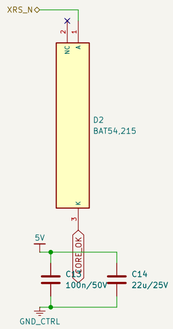


Supply provenance (PWR_FLAG)

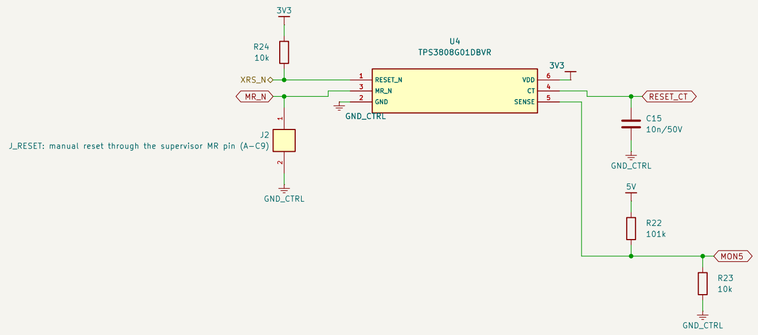








TPS3808 5 V supervisor and reset





AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /5 V buck, dual LDO, supervisor and reset/
File: 02_supplies.kicad_sch

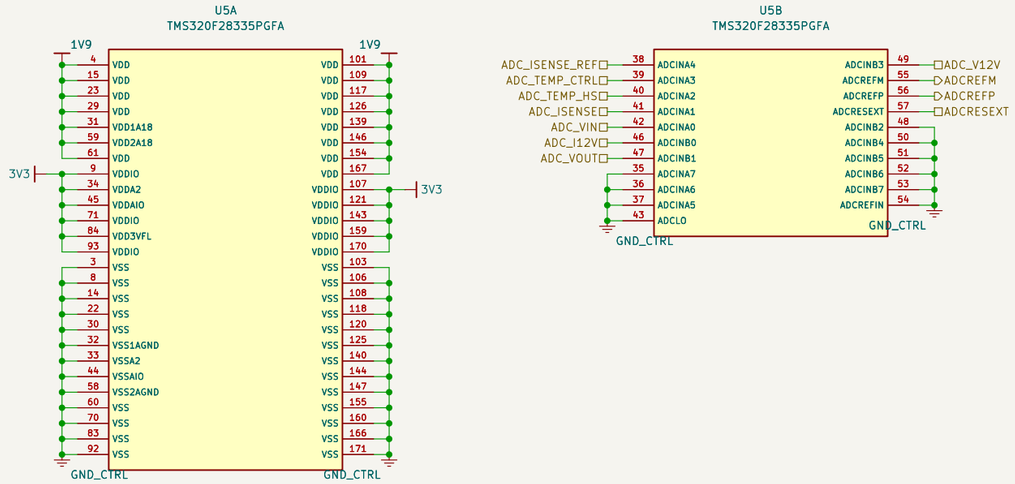
Title: 5 V buck, dual LDO, supervisor and reset

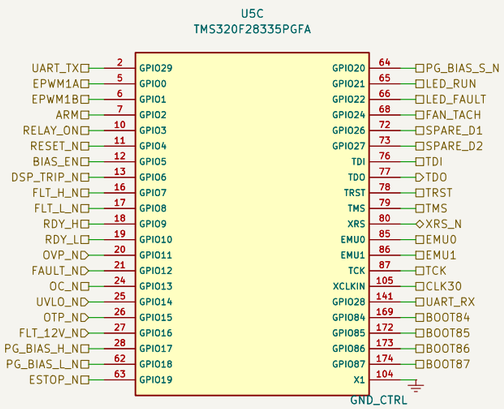
Size: A2 | Date: | **Rev: V2.2**
KiCad E.D.A. 10.0.7 | Id: 3/9

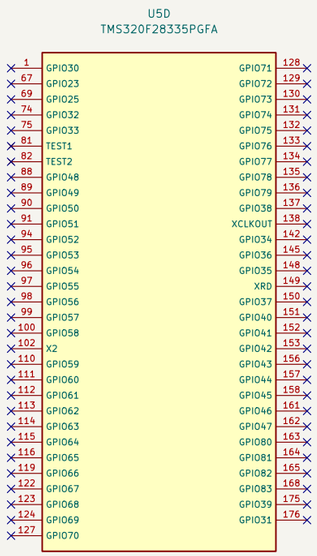
TMS320F28335 — all 176 pins

TMS320F28335PGFA, all 176 pins; units split into supplies, ADC, connected GPIO and unused pins. Pin map per design_decisions.json.

TMS320F28335PGFA







AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /TMS320F28335 — all 176 pins/
File: 03_dsp.kicad_sch

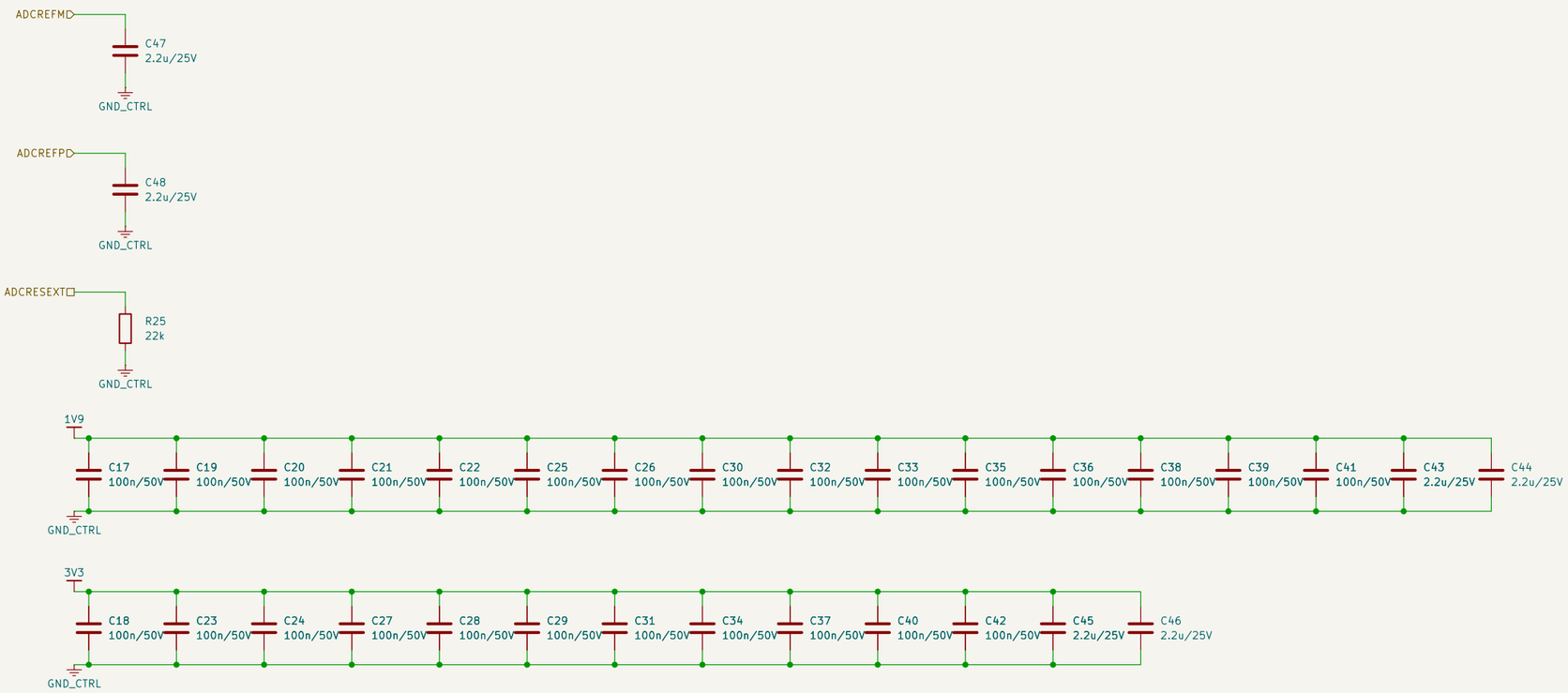
Title: TMS320F28335 — all 176 pins

Size: A3	Date:	Rev: V2.2
KiCad E.D.A. 10.0.7		Id: 4/9

DSP per-pin bypass and ADC reference

One 100 nF per DSP supply pin (26), 2 x 2.2 uF per rail, 2.2 uF on ADCREFP/ADCREFM, 22 k on ADCRESEXT (SPRS439Q).

DSP per-pin bypass and ADC reference



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /DSP per-pin bypass and ADC reference/
File: 04_dsp_decoupling.kicad_sch

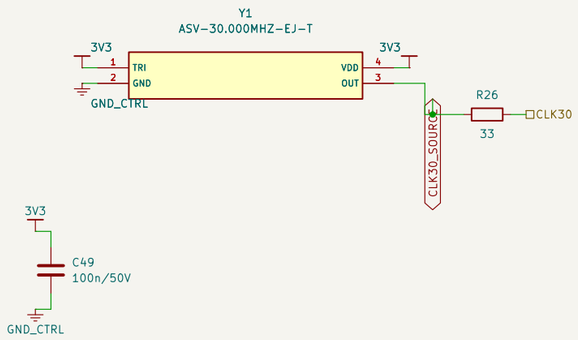
Title: DSP per-pin bypass and ADC reference

Size: A3	Date:	**Rev: V2.2**
KiCad E.D.A. 10.0.7		Id: 5/9

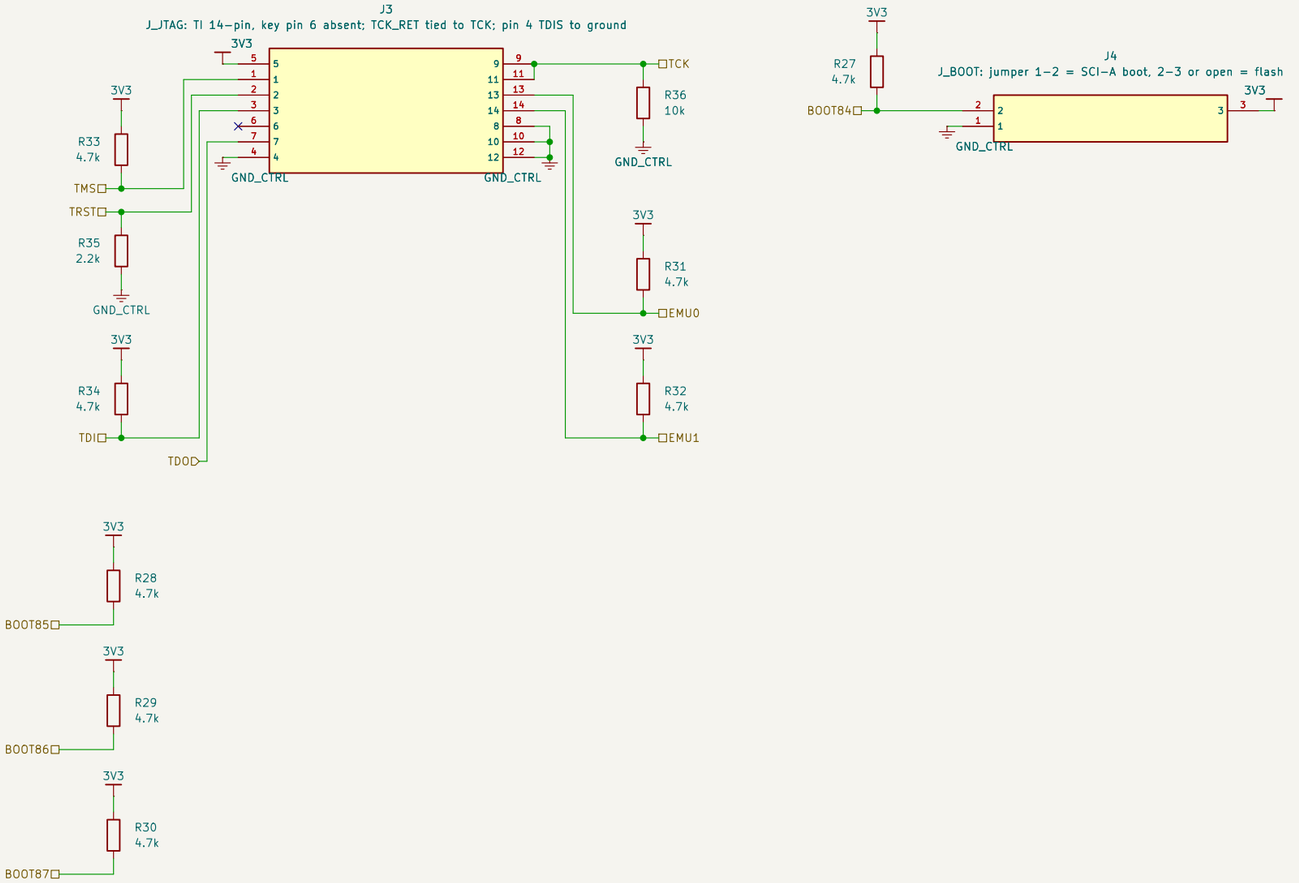
Oscillator, boot straps, JTAG, UART

30 MHz oscillator on XCLKIN (X1 grounded, X2 open), 33 ohm series; boot straps GPIO84–87 pulled high; TI 14-pin JTAG; SCI-A header.

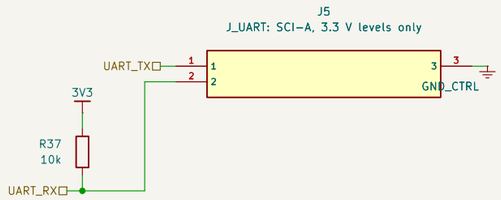
30 MHz oscillator



Boot straps and JTAG



SCI-A UART header



AI-GENERATED ENGINEERING PROTOTYPE – V2.2 – HUMAN REVIEW REQUIRED

Sheet: /Oscillator, boot straps, JTAG, UART/
File: 05_clock_debug.kicad_sch

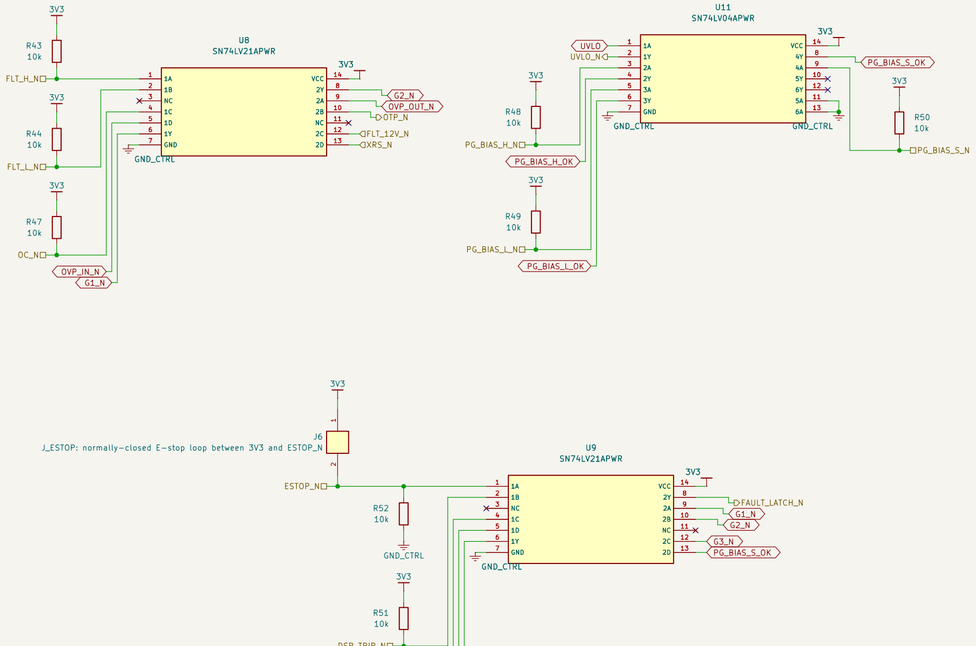
Title: Oscillator, boot straps, JTAG, UART

Size: A3 | Date: | **Rev: V2.2**
KiCad E.D.A. 10.0.7 | Id: 6/9

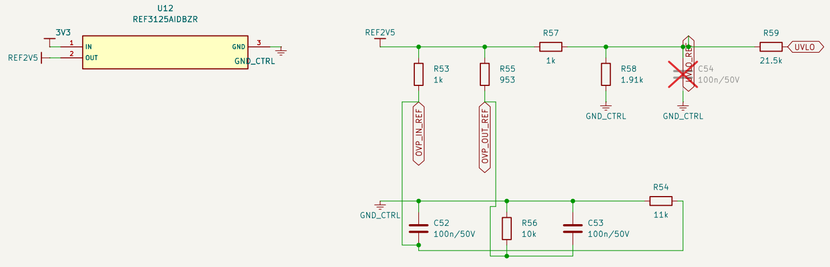
PWM gating, fault latch, combiners, comparators

ePWM1A/1B AND GATE_EN (SN74LVC2G08) -> 33 ohm -> mezzanine; SN74LVC1G74 fault latch (D = 1, CLK = ARM, CLR_N = FAULT_LATCH_N, Q = FAULT_N = ARMED); 3 x SN74LV21A combiners; SN74LV04A restores the UVLO and PG_BIAS polarities (A–C6, A–C7); TLV1702 comparators with REF3125-derived thresholds.

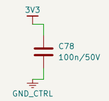
Fault combiners, polarity restoration, pull-ups, E-stop



REF3125 reference and OVP / UVLO thresholds



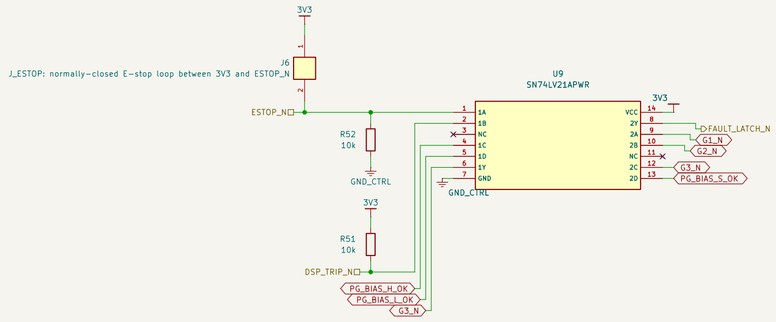
U10 dedicated local bypass (V2.2)



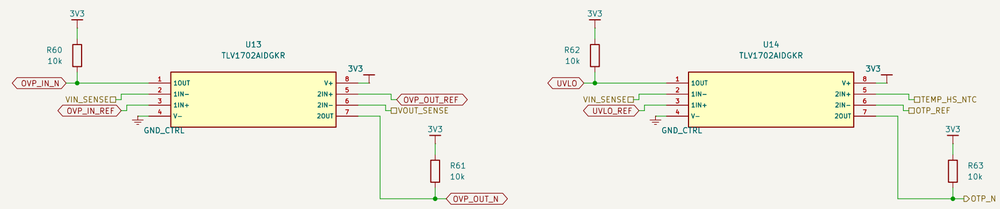
U11 dedicated local bypass (V2.2)



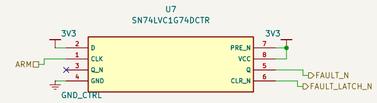




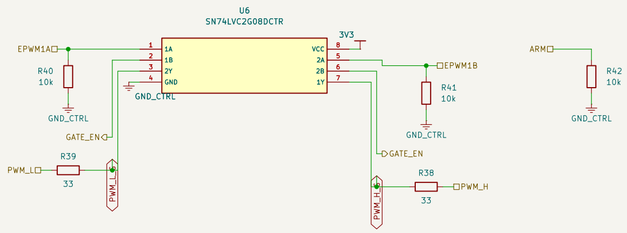
TLV1702 comparators



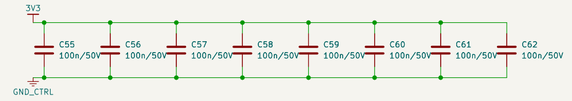
Fault latch (SN74LVC1G74)



PWM gating (SN74LVC2G08)



Logic supply bypass



U8 dedicated local bypass (V2.2)



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /PWM gating, fault latch, combiners, comparators/
File: 06_pwm_fault_logic.kicad_sch

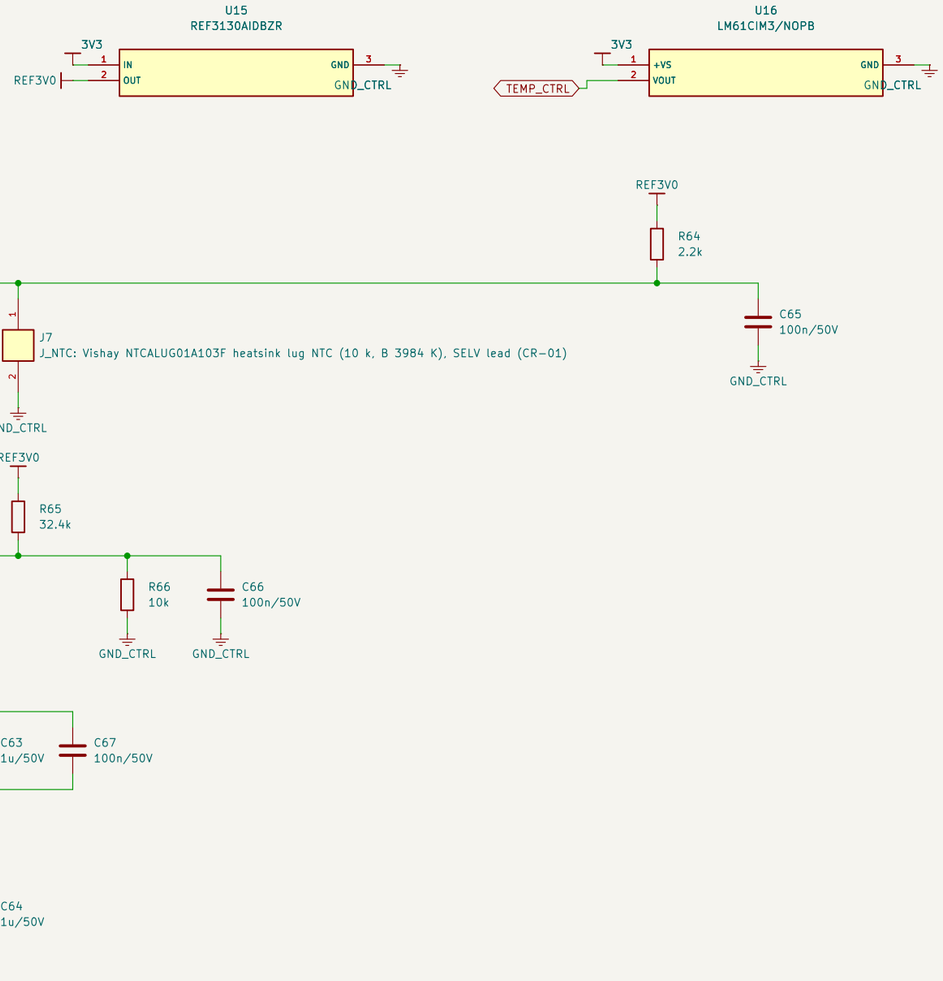
Title: PWM gating, fault latch, combiners, comparators

Size: A2	Date:	Rev: V2.2
KiCad E.D.A. 10.0.7		Id: 7/9

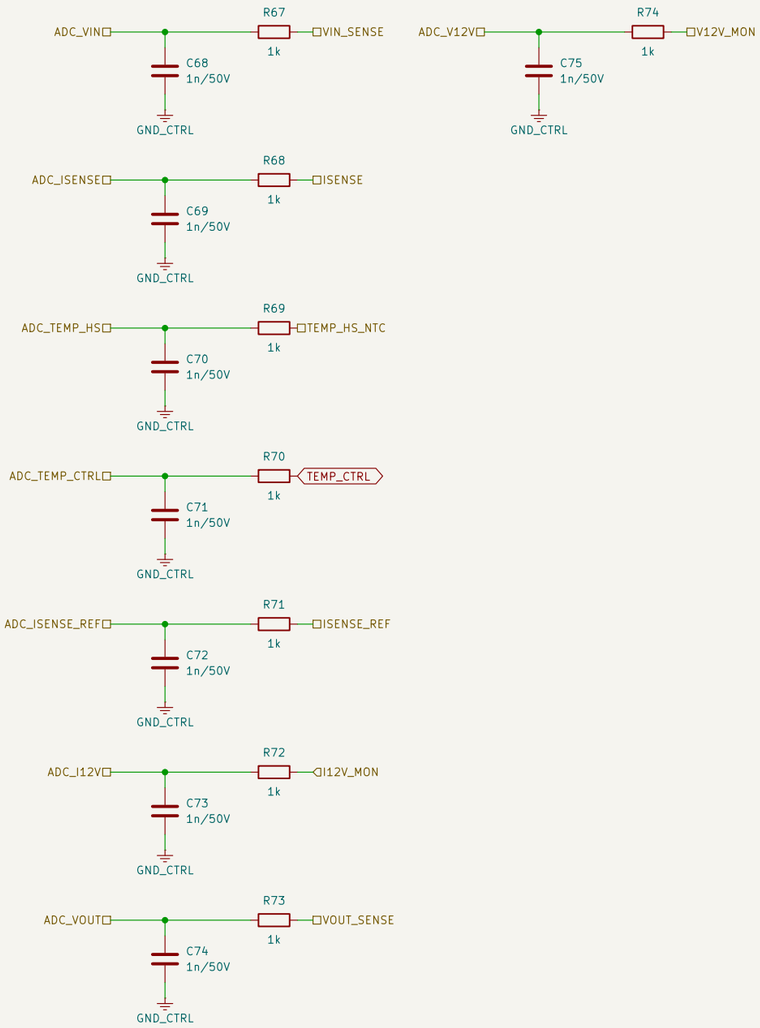
ADC conditioning, NTC readout, monitors

1 k + 1 nF at every ADC input; heatsink NTC with 2.2 k to REF3V0 (REF3130); OTP threshold 42.2 k / 10 k = 0.575 V (110 C); LM61.

REF3130, heatsink NTC, OTP threshold, LM61



ADC anti-alias RC networks



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /ADC conditioning, NTC readout, monitors/
File: 07_adc_frontend.kicad_sch

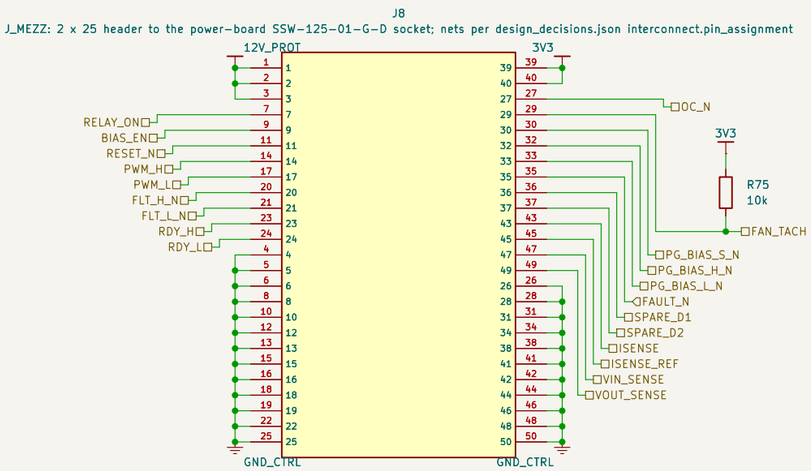
Title: ADC conditioning, NTC readout, monitors

Size: A3	Date:	**Rev: V2.2**
KiCad E.D.A. 10.0.7		Id: 8/9

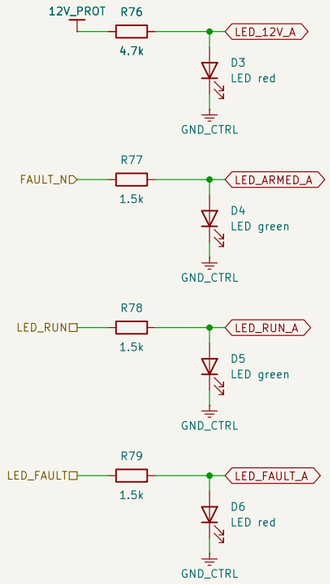
Mezzanine header, connectors, LEDs, test points, mounting

Mezzanine header TSW-125-07-G-D (pin assignment per design_decisions.json), indicator LEDs, test pads, 4 x M3.

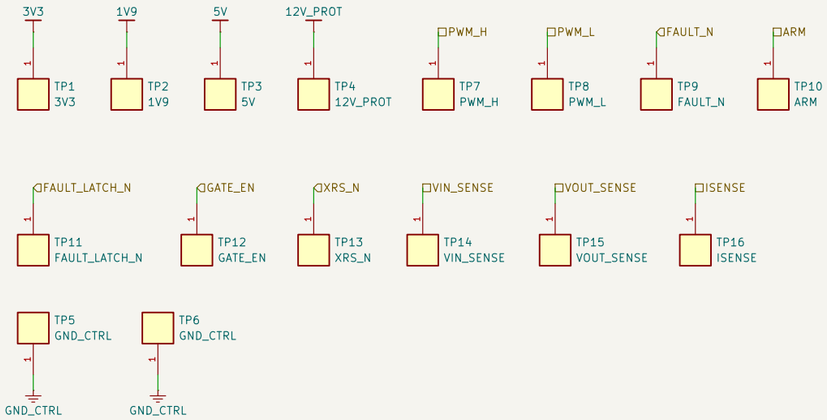
Mezzanine header to the power board



Indicator LEDs



Test points



Mechanical



AI-GENERATED ENGINEERING PROTOTYPE — V2.2 — HUMAN REVIEW REQUIRED

Sheet: /Mezzanine header, connectors, LEDs, test points, mounting/
File: 08_interconnect.kicad_sch

Title: Mezzanine header, connectors, LEDs, test points, mounting

Size: A3	Date:	**Rev: V2.2**
KiCad E.D.A. 10.0.7		Id: 9/9